Research Methodology

Week 11 Assignment: Questions, Answers and Explanations

Week 11: The Role of AI in Research: Introduction to Generative AI

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

The portal records no submission from you for this week, so it shows 0 on every question. The correct answers below are the portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. These questions are based on the Introduction to Generative AI lecture.

Answer key at a glance

Question	Type	Correct option(s)	Portal score
1	Select all that apply	**B, C**	Not submitted
2	Select all that apply	**B, C**	Not submitted
3	Select all that apply	**A, C, D**	Not submitted
4	Select all that apply	**A, B, D**	Not submitted
5	Select all that apply	**A, B, C**	Not submitted
6	Select all that apply	**A, B, D**	Not submitted
7	Select all that apply	**A, B, D**	Not submitted
8	Select all that apply	**A, B, D**	Not submitted
9	Select all that apply	**A, B, C**	Not submitted
10	Select all that apply	**B, C, D**	Not submitted

Questions with explanations

Question 1 of 10

The lecture uses the example of Arthur Samuel's checkers program to illustrate an early form of learning by machines. Which of the following statements are correct?

Select all that apply.

Option	Statement	Verdict	Reason
A	The machine was explicitly programmed with every possible checkers strategy.	**Wrong**	The program improved by playing; it was not given every strategy.
B	**The machine learned through experience by repeatedly playing the game.**	**Correct**	Learning from experience is the point of the example.
C	**The example is associated with an early instance of reinforcement learning.**	**Correct**	Improving from the outcomes of games is reinforcement learning.
D	The example demonstrates that AI-based learning is a phenomenon that began only with generative AI.	**Wrong**	The program dates from the 1950s, long before generative AI.

Correct answer: B and C.

Explanation: Samuel's program improved by playing many games and adjusting from the outcomes. This is learning from experience and an early form of reinforcement learning, and it shows that machine learning is much older than generative AI.

Question 2 of 10

Which of the following statements correctly reflect the lecture's discussion of classical AI and generative AI?

Select all that apply.

Option	Statement	Verdict	Reason
A	Generative AI is the only appropriate approach for solving modern research problems.	Wrong	"The only" is too strong; the right tool depends on the problem.
B	**Classical AI methods can still provide effective solutions for domain-specific problems.**	Correct	Classical methods are often better and cheaper for narrow, well-defined tasks.
C	**Researchers should select an AI approach according to the problem rather than assuming that generative AI must always be used.**	Correct	The tool should fit the problem.
D	Classical AI became irrelevant after the emergence of LLMs.	Wrong	Classical AI is still useful.

Correct answer: B and C.

Explanation: Choose the approach according to the problem. Classical AI methods remain effective for domain-specific tasks, and generative AI is not always the right choice.

Question 3 of 10

Consider an LLM that predicts the next token from the context provided to it. Which of the following statements are correct?

Select all that apply.

Option	Statement	Verdict	Reason
A	**The probability of a token can depend on the preceding context.**	Correct	The model assigns probabilities given the earlier text.
B	A token with the highest probability must always be generated.	Wrong	The model can sample from the distribution, so the top token is not always chosen.
C	**Different contexts can result in different probability distributions for the next token.**	Correct	A new context changes the probabilities.
D	**The model can repeatedly predict tokens to construct a longer response.**	Correct	Each predicted token is added to the text and the process repeats.

Correct answer: A and C and D.

Explanation: A token is a word or word-piece. The model gives each possible next token a probability that depends on the context, picks one, appends it and repeats, which builds a long answer. The most probable token is not forced, which is why answers can vary.

Question 4 of 10

Which of the following correctly distinguish token size, context size, and parameter count in an LLM?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Token size is related to the vocabulary of words or subwords that the model can generate.**	**Correct**	The vocabulary sets the pieces of text the model can produce.
B	**Context size refers to the amount of previous token information considered by the model.**	**Correct**	It is how much earlier text the model can take into account.
C	Parameter count refers to the number of tokens that can be stored in the prompt.	**Wrong**	The number of tokens in a prompt is the context size, not the parameter count.
D	**Parameter count is related to how the learned probability distributions are represented in the model.**	**Correct**	Parameters are the learned numbers that encode the probability patterns.

Correct answer: A and B and D.

Explanation: Vocabulary sets what the model can output, context size is how much text it can consider at once, and parameters are the learned numbers that hold what it has learned.

Question 5 of 10

Which statements correctly describe the role of vector representations in the LLM framework explained in the lecture?

Select all that apply.

Option	Statement	Verdict	Reason
A	**A very large probability matrix can be represented through a comparatively smaller-dimensional vector space.**	**Correct**	Storing a probability for every possible context is impossible, so contexts are compressed into vectors.
B	**Different contexts may be projected to nearby locations in the vector space.**	**Correct**	Similar contexts end up close together.
C	**Such compression can contribute to variability and errors because information from different contexts may influence generation.**	**Correct**	The sharing is useful but lets one context affect another.
D	Every possible context is retained as a completely independent point with no overlap.	**Wrong**	That would mean no compression at all.

Correct answer: A and B and C.

Explanation: Contexts are compressed into vectors of smaller dimension. Similar contexts sit near each other, which is efficient but also lets one context influence another and causes errors.

Question 6 of 10

Which of the following statements are consistent with the lecture's explanation of alignment?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Alignment can modify model behaviour so that generated probabilities better correspond to desired outcomes.**	Correct	Alignment adjusts the model so its outputs match what people want.
B	**Alignment may be directed towards objectives such as factuality, bias reduction or local context.**	Correct	These are typical alignment goals.
C	Alignment guarantees that all outputs produced by an LLM are factually correct.	Wrong	It reduces errors but cannot guarantee correctness.
D	**Reinforcement learning from human preferences is associated with the alignment process.**	Correct	Learning from human feedback is a common alignment method.

Correct answer: A and B and D.

Explanation: Alignment adjusts the model after basic training so that outputs match desired outcomes such as factuality, less bias or local context, often using human feedback. It cannot guarantee that every output is correct.

Question 7 of 10

A researcher asks a small language model a question whose meaning is ambiguous. Relevant information is retrieved from an internal repository and added to the original question before generating the answer. Which statements correctly describe this approach?

Select all that apply.

Option	Statement	Verdict	Reason
A	**The approach is consistent with Retrieval-Augmented Generation (RAG).**	Correct	RAG means fetching relevant material and adding it to the prompt.
B	**The retrieved information provides additional context to the model.**	Correct	The extra text is the added context.
C	The approach necessarily requires retraining the entire language model.	Wrong	Only the input changes; the model is not retrained.
D	**Additional context can make the model's response more focused.**	Correct	More context narrows what the question means.

Correct answer: A and B and D.

Explanation: RAG retrieves relevant information and puts it into the prompt, so the model has more context and can answer more precisely. The model itself is not retrained.

Question 8 of 10

The lecture uses the example of the 2011 ICC World Cup to explain the effect of retrieval-augmented generation. Which of the following correctly captures the idea?

Select all that apply.

Option	Statement	Verdict	Reason
A	**Additional retrieved information can contextualize an otherwise ambiguous query.**	**Correct**	Extra facts remove the ambiguity.
B	**The original question can be effectively shifted toward a more appropriate contextual representation.**	**Correct**	The added context moves the question to a better point in the vector space.
C	RAG works primarily by increasing the number of parameters in the LLM.	**Wrong**	RAG does not change the number of parameters.
D	**A richer context can increase the probability of producing the intended (cut off by the portal)**	**Correct**	Better context makes the intended answer more probable.

Correct answer: A and B and D.

Explanation: A short question such as "who won the World Cup" is ambiguous. Adding the year and event gives context that moves the question to where the intended answer is more probable. RAG changes the input, not the model.

Note: *The portal cuts off the end of option D ("...producing the intended"); it presumably continues "answer". The answer key counts it as correct.*

Question 9 of 10

Which of the following statements about hallucination are correct according to the lecture?

Select all that apply.

Option	Statement	Verdict	Reason
A	**A model may produce a plausible answer even when the generated information does not match the ground truth.**	**Correct**	The model produces the most probable continuation, which can sound right and be false.
B	**Hallucination can occur when the model has insufficiently represented information or context.**	**Correct**	Little information about a topic makes errors more likely.
C	**Hallucination can be reduced through techniques such as providing larger context, but it cannot necessarily be eliminated completely.**	**Correct**	More context helps but cannot remove the problem.
D	Hallucination means that the model has deliberately decided to provide false information.	**Wrong**	The model has no intention to deceive.

Correct answer: A and B and C.

Explanation: Hallucination is a plausible but wrong output, more likely when information or context is thin. It can be reduced but not guaranteed away, and it is not deliberate lying.

Question 10 of 10

A language model produces a misleading recommendation because a false or deliberately misleading article was present in information from which the model learned. Which statements are appropriate?

Select all that apply.

Option	Statement	Verdict	Reason
A	The problem is necessarily an example of hallucination.	**Wrong**	Here the model repeated a falsehood from its sources; it did not invent a fact.
B	**AI systems can potentially propagate misinformation present in their information sources.**	**Correct**	A model can repeat falsehoods it learned.
C	**A response may be misleading even when it is not simply an invented fact by the model.**	**Correct**	Misleading answers can come from the sources.
D	**Independent verification of important AI-generated information remains necessary.**	**Correct**	Because either cause is possible, important claims should be checked.

Correct answer: B and C and D.

Explanation: A model can repeat misinformation from its sources, which is different from inventing a fact (hallucination). Either way, important AI output should be verified independently.